

MD SADMAN SIRAJ

PH.D. IN COMPUTER ENGINEERING, SPECIALIZATION ON APPLIED AI FOR CYBER-PHYSICAL SYSTEMS

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PROFESSIONAL SUMMARY

- 4 years of Ph.D. research experience focusing on developing distributed decision-making algorithms using Applied Artificial Intelligence and Game Theory for alternative localizations, resource management and cybersecurity for wireless cyber-physical systems
- Designed and deployed a wireless testbed enabling multi-node wireless networking experiments for HELIO-COMM research project (U.S. DOE), developing skills in wireless communications and embedded systems
- Built an ML pipeline to detect malicious modifications in wireless packets for Goaltender research project (U.S. DOE), gaining expertise in machine learning and deep learning
- Developed contract-theory-based trust models and evaluated them on 20 FPGA ZYBO-Z7-10 boards with PUFs for CBDC research project (Bank of Canada), advancing expertise in cybersecurity and hardware trust
- Strong and consistent record of publications throughout the Ph.D. program

EDUCATION

- **Ph.D., Computer Engineering** May 2026
Arizona State University, USA CGPA 4.17

Thesis Title: *Intelligent Decision-making for Complex and Dynamic Wireless Cyber-physical Systems*

Advisor: Dr. Eirini Eleni Tsiropoulou, Associate Professor, ECEE, ASU

Research Interests: Wireless Cyber-physical Systems, Applied Artificial Intelligence including Supervised, Unsupervised, Reinforcement and Federated Learning, Alternative Localizations, and Distributed Energy Resources (DERs)

- **M.Sc., Computer Engineering (with Distinction)** December 2023
University of New Mexico, USA CGPA 4.21

Thesis Title: *A Bio-inspired Alternative Positioning, Navigation, and Timing Approach based on a Potential Game-theoretic Model*

Advisor: Dr. Eirini Eleni Tsiropoulou, Associate Professor, ECE, UNM

SKILLS

- **Programming Languages:** Python, MATLAB, C, C++, SQL
- **Hardware/Embedded Programming:** Firmware Development, Zephyr-RTOS, Nordic Semiconductor's nRF7002 Wi-Fi Chip and nRF5340 System-on-Chip (SoC), FPGA ZYBO Z7-10, ADALM-PLUTO SDR, USRP B210 SDR
- **Protocols and Tools:** I2C, SPI, UART, USB, TCP/IP, RF, Wi-Fi, Zigbee, Bluetooth Low Energy (BLE), BLE Mesh, Enhanced ShockBurst (ESB), IEEE 802.11/802.15.4 wireless protocols, DSP in MATLAB, Scikit-learn, Tensorflow, PyTorch, Pandas, SciPy, HPC, Linux, SLURM, Git, Docker, AWS (EC2)
- **Software Development Tools:** nRF SDK, VS Code, GNU Radio, NS3, Android Studio, Eclipse
- **Other skills:** Research and open data aggregation, data cleaning and processing, parallel processing, excellent visualizations, collaborative project management, advanced presentation skills

EXPERIENCE

- **Graduate Research Associate, Arizona State University** January 2025 – May 2026
Performance and Resource Optimization in Networks Laboratory Continued Attendance from UNM

- Trained AI models to detect malicious modifications in wireless packets for electric vehicle charging networks.
- Developed contract theory-based trust models for hardware-oriented cybersecurity in digital currency exchange.
- Deployed a wireless testbed enabling multi-node wireless communication with low-latency wireless traffic.

- **Graduate Research Assistant, University of New Mexico** January 2022 – December 2024
Performance and Resource Optimization in Networks Laboratory Transferred to ASU

- Developed simulation frameworks to model and evaluate distributed decision-making algorithms for complex cyber-physical systems using artificial intelligence and game theory for solving alternative localization challenges in GPS/GNSS-denial scenarios and resource allocation problems in wireless networks.

RESEARCH PROJECTS

- **GOALTENDER Research Project (funded by U. S. Department of Energy)** January 2024 – May 2026
Collaborated with a team from Sandia National Laboratories (SNL)

- Explored ML and DL models for anomaly detection in electric vehicle (EV) charging networks including supervised methods (Random Forest, XGBoost), unsupervised methods (Isolation Forest, One-Class SVM), and deep learning (Autoencoders).

- Detected stealthy under-billing, over-billing, and spoofing state of charge attacks using AI with 97.9%, 80.84% and 94% F1 score, respectively.
- Integrated Large Language Models (LLMs - GPT-4.0, Llama-3.1 80B and Llama-3.3 70B) to generate human-readable root cause explanations for detected anomalies, improving explainability and operational insights.

PEERTRUST Research Project (funded by Bank of Canada) December 2023 – May 2026

- Built a secured and trusted digital currency exchange system using reinforcement learning and Physical Unclonable Function (PUF) for peer-based authentication which can be effectively used in distributed and offline configurations, eliminating the need for persistent connectivity to a centralized authority.

HELIOCOMM Research Project (funded by U. S. Department of Energy) April 2023 – May 2026

Collaborated with a team from National Laboratory of the Rockies (NLR)

- Designed and modeled a resilient wireless communication system using IEEE 802.11/802.15.4 wireless protocols and reinforcement learning in Python, introducing the first wireless network system design and research in the USA for heliostat fields.
- Validated the efficiency and reliability of the wireless network design with large-scale system performance simulations in HPC environments, manifesting very low communication delays ranging from 250 millisecond to 2 second.

Range Extension in Wi-Fi Networks for Terrain Classification

Core Idea: Dynamic switching between the Station (STA) and Soft-AP (SAP) modes in nRF7002 Wi-Fi chips

- Terrain data collection from wireless nodes operating in Wi-Fi STA mode and communicating with a central server for Machine Learning model training and prediction.
- Nodes switching to SAP mode to provide Access Point (AP) services to nodes outside the typical network coverage or communication range.
- Experimental verification of Wi-Fi network range extension beyond typical coverage in outdoor environment.
- Software or application update through Wi-Fi and UDP-based Firmware Update Over the Air (FOTA).

Multi-node Support in Large-scale IoT Networks for Industrial Monitoring

Core Idea: Co-existence of Wi-Fi and ESB using nRF7002 Wi-Fi chip and nRF5340 System-on-Chip (SoC)

- Large-scale industrial monitoring through a densely deployed industrial IoT network with multi-node support through Wi-Fi and ESB protocols.
- Custom ESB collision avoidance and multiplexing schemes for multi-node simultaneous transmissions.
- Performance evaluation through hardware-based experiments revealing low latency (500 ms) and low packet drop rates (0.01%).
- nRF5340 SoC network core image update supported by PCD (Peripheral CPU DFU) triggered by the application core through Wi-Fi and MQTT-based Firmware Update Over the Air (FOTA).

NOTABLE PUBLICATIONS AND PRESENTATIONS

Google Scholar: scholar.google.com/citations?user=S5HhON8AAAAJ

Technical Papers

- **M. S. Siraj**, A. B. Rahman, M. Diamanti, E. E. Tsiropoulou, and S. Papavassiliou, "Alternative positioning, navigation, and timing enabled by games in satisfaction form and reconfigurable intelligent surfaces," *IEEE Systems Journal*, vol. 17, no. 3, pp. 5035-5046, 2023.
- **M. S. Siraj**, E. E. Tsiropoulou, S. Papavassiliou, and J. Plusquellic, "Symbiotic positioning, navigation, and timing via game theory and reinforcement learning," *IEEE Access*, vol. 13, pp. 69532-69546, 2025.
- **M. S. Siraj**, P. Charatsaris, M. Diamanti, E. E. Tsiropoulou, and S. Papavassiliou, "Heroes: Humanitarian emergency response based on uav-enabled integrated sensing and communication, positioning, and satisfaction games," *ACM J. Auton. Transport. Syst.*, vol. 2, no. 4, Jun. 2025.
- **M. S. Siraj**, J. R. Atencio, and E. E. Tsiropoulou, "Panther: A power-optimized and accurate positioning, navigation, and timing with high efficiency and reliability," *IEEE Open Journal of the Communications Society*, vol. 6, pp. 3265-3279, 2025.
- **M. S. Siraj**, A. B. Rahman, and E. E. Tsiropoulou, "Wireless control in commercial concentrated solar power fields," *Solar Energy*, vol. 306, p. 114289, 2026.
- **M. S. Siraj**, A. B. Rahman, C. Minwalla, E. E. Tsiropoulou, and J. Plusquellic, "Sourcing Trust From Peers with Physical Unclonable Functions," in *2025 IEEE International Symposium on Hardware Oriented Security and Trust (HOST)*, San Jose, CA, USA, 2025, pp. 268-278.

HONORS AND AWARDS

University Graduate Fellowship	2026
<i>Arizona State University</i>	<i>Tempe, AZ, USA</i>
ECCE Graduate Research Symposium Award	2025
<i>Arizona State University</i>	<i>Tempe, AZ, USA</i>
IEEE Service Award 2024	2024
<i>IEEE Albuquerque Section</i>	<i>Albuquerque, NM, USA</i>
IEEE Outstanding Graduate Engineering Student Award 2023	2023
<i>IEEE Albuquerque Section</i>	<i>Albuquerque, NM, USA</i>
ECE Outstanding Student Teaching Award 2023	2023
<i>Department of Electrical and Computer Engineering, University of New Mexico</i>	<i>Albuquerque, NM, USA</i>